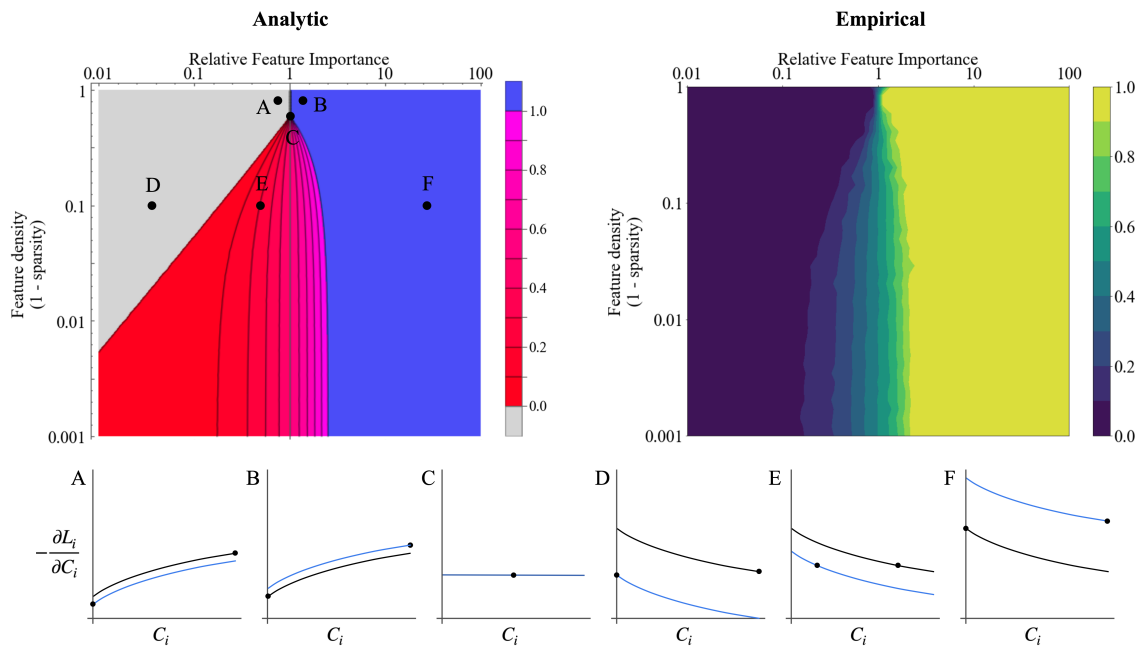

Figure 4: Upper: Analytical and empirical phase diagrams for our toy model with 6 features and 3 neurons. In both panels one feature has a different importance from the rest, and colors show the resulting capacity allocation for that feature as a function of sparsity and relative importance. Lower: Plots of marginal loss reduction $\partial L/\partial C_i$ as a function of feature capacity for each labelled point in the analytical phase diagram. The blue curve represents the feature with varied importance and the black one represents the constant important feature. Black dots are optimal allocations of capacity.

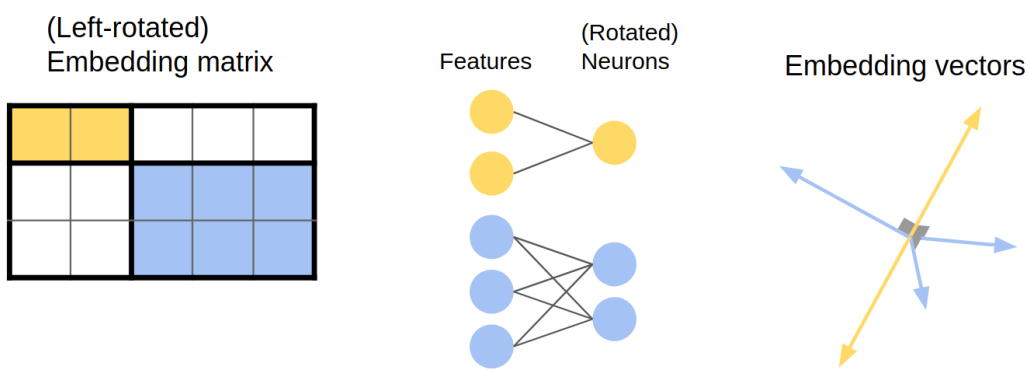

Figure 5: Left: An embedding matrix with two blocks. Center: The relationship between features and (principal-component-aligned) neurons for this matrix. Right: Embedding vector geometry for this matrix.

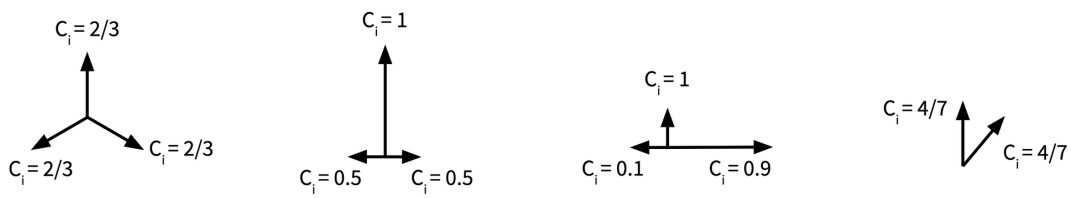

Figure 6: Example capacity allocations for different embeddings.

$$e \equiv W \cdot x \tag{1}$$
$$h \equiv \mathrm{ReLU}(e) \tag{2}$$

with $W \in \mathbb{R}^{d \times p}$, $x \in \mathbb{R}^p$, $e, h \in \mathbb{R}^d$. We associate each dimension of the input vector x with a feature, and we call each dimension of the non-linear layer a neuron.

For simplicity, in the rest of this paper we work with a one-layer model, but our capacity definition should be valid for any layer in a multi-layer model.

When a model represents a feature in the input space, it is convenient to think that it expends some capacity to do so. Our intuition here is that as we ask a model to represent more and more features we eventually exhaust its ability to do so, resulting in features interfering. We study the superposition phenomena by asking the question: "How do models allocate limited representation capacity to input features?" In what follows we assume that each input feature is assigned a unique dimension in the input space (e.g. feature i is input dimension i), and capacity we define below.

Let $W_{\cdot,i} \in \mathbb{R}^d$ be the embedding vector for feature i. The capacity allocated to feature i is

$$C_i = \frac{(W_{\cdot,i} \cdot W_{\cdot,i})^2}{\sum_j (W_{\cdot,i} \cdot W_{\cdot,j})^2} \tag{3}$$

We can think of C_i as "the fraction of a dimension" allocated to feature i ([EHO+22b])[1]. The numerator measures the size of the embedding and the denominator tracks the interference from other features. By this definition, C_i is bounded between 0 and 1. In the case $W_{\cdot,i} = 0$, where this expression is undefined, we set $C_i = 0$.[2]

We define the total model capacity to be $C = \sum_i C_i$ (in a multi-layer model, this would be a single layer-pair's capacity). This is bounded between 1 and the embedding dimension D (see Appendix F for a proof of the upper bound).

Note that a set of capacities does not uniquely specify a weight matrix. For example, capacity is invariant to the overall scaling and rotation of W. In what follows it will be useful to have a full parameterization of W that includes C_i, so we define S to be a set of additional parameters that uniquely specify a weight matrix W given its capacities $C_1, \ldots, C_N$. We can then parameterize the loss using $(C_1, \ldots, C_N, S)$ rather than W.

2.2 Loss Minimization

We are interested in how loss minimization allocates capacity among different features. Because the capacity of each feature lies in $[0, 1]$ and there is also a constraint on the total capacity of a model, this is a constrained optimization problem:

$$\begin{aligned} \min_{C_{1:n}, S} \quad & L(C_{1:n}, S) \\ \text{s.t.} \quad & 0 \leq C_i \leq 1 \\ & 1 \leq \sum_i C_i \leq D \end{aligned}$$

[1]We can also interpret C_i as the squared correlation coefficient between x_i and $(W^T W x)_i$ – see Appendix E.

[2]This is the limit of the expression from almost all directions, so in practice $W_{\cdot,i} \approx 0$ implies $C_i \approx 0$.

In brief, we minimize over S first to find the $S^*(C_i)$ minimizing the loss[3]. We then minimize over C_i Intuitively capacity should always be a good thing, so we are interested in situations where the loss function L is monotonically decreasing in C_i. In this case, the bound on the total capacity $\sum_i C_i \leq D$ will be saturated, and we can replace this constraint with a Lagrange multiplier:

$$\begin{aligned} \min_{C_{1:n},\lambda} \quad & L(C_{1:n}) + \lambda \left(\sum_i C_i - D \right) \\ \text{s.t.} \quad & 0 \leq C_i \leq 1 \end{aligned} \tag{4}$$

Because we have independent constraints on each C_i, for each i the minimum occurs either at a boundary ($C_i = 0$ or 1) or in between at a point where

$$\frac{\partial}{\partial C_i} \left(L(C_{1:n}) + \lambda \left(\sum_i C_i - D \right) \right) = 0 \tag{5}$$

The derivative in the three cases will obey

$$\frac{\partial L(C_{1:n})}{\partial C_i} \geq -\lambda \quad (C_i = 0) \tag{6}$$

$$\frac{\partial L(C_{1:n})}{\partial C_i} = -\lambda \quad (0 < C_i < 1) \tag{7}$$

$$\frac{\partial L(C_{1:n})}{\partial C_i} \leq -\lambda \quad (C_i = 1) \tag{8}$$

In other words, at the minimum of the loss, capacity is allocated to each feature until the marginal value (decrease in loss) from additional capacity is at a constant threshold – unless the feature is so important that the marginal value is above threshold even when fully represented, or so unimportant that the marginal value is below threshold even when the feature is ignored.

Note that none of this derivation depended on the precise definition of capacity. We only had to assume that (1) allocating more capacity to a feature never makes the loss greater, (2) all capacity allocations satisfying the feature-level and total constraints are realizable by some embedding matrix. Thus we could have used other definitions, and indeed in Appendix G we explore one such alternative.

We can visualize this in terms of the graph of $\partial L/\partial C_i$ as a function of C_i, holding C_j constant for $j \neq i$. For instance, in Figure 2 there are diminishing returns: $\partial^2 L/\partial C_i^2 > 0$, so that each additional unit of capacity makes the next one less useful. If $\partial L/\partial C_i$ crosses $-\lambda$, the value of C_i where it crosses will be the optimal allocation. Features only get 0 or 1 capacity if the entire $\partial L/\partial C_i$ curve is above or below $-\lambda$ (respectively).

By contrast, in the case of increasing returns ($\partial^2 L/\partial C_i^2 < 0$), every feature will have $C_i = 0$ or $C_i = 1$ and superposition will not occur at all. As we show below for several different toy models, diminishing returns occur for inputs of high sparsity or kurtosis, and accelerating return for inputs of low sparsity or kurtosis.

3 Quadratic model

We now instantiate our theory for a two-layer model with quadratic activations. This choice of activation function allows us to study this model analytically in some detail.

We define our model in Section 3.1. In Section 3.2 we analyze the loss function for the case of independent random input features. We are able to write the expected loss entirely in terms of the embedding lengths and the variance and kurtosis of feature strengths.

In Section 3.3 we demonstrate a surprising feature of this quadratic model: the loss function can be written in terms of the feature capacity and the embedding lengths. This allows us to analytically explore the marginal benefit of allocating more capacity to each feature.

We then go on to solve for the optimum allocation of capacity across features (Section 3.4) and relate this to our results from Section 2).

[3]There are multiple ways to define S, which give different $(\partial L/\partial C_i)$ holding "S" constant. Minimizing over S removes this ambiguity.